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Article

Interpretable Deep Learning Framework Enables Authentication and Quantification of Olive Oil Adulteration by Raman Spectroscopy

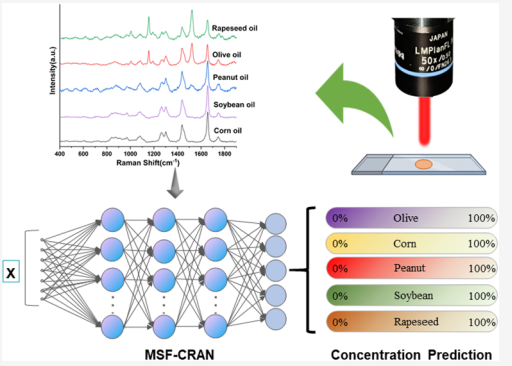
Xue-Yang Xiong, Huan Chen, Jia-Sheng Lin,* and Fan-Li Zhang*

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ABSTRACT: The high economic value of extra virgin olive oil (EVOO) makes it susceptible to adulteration with lower-cost oils, necessitating rapid and reliable analytical methods for authenticity assessment. In this work, a Raman spectroscopic approach combined with a multiscale Fusion Convolutional Residual Attention Network (MSF-CRAN) is developed for the identification and quantification of EVOO adulteration. Model interpretability is enhanced through the integration of SHapley Additive Explanations (SHAP), enabling the analysis of spectral feature contributions. To address limited sample availability in multicomponent adulteration systems, binary, ternary, and higher-order blended samples were prepared, and data set expansion was achieved using spectral feature transfer and generative adversarial networks (GANs). The proposed MSF-CRAN model demonstrates excellent performance, achieving 100% classification accuracy for adulteration identification and high quantitative accuracy for ternary mixtures with an R^2 of 0.991 and a mean absolute error (MAE) of 0.0181. The results highlight the robustness and generalization capability of the proposed framework for the Raman-based analysis of complex multicomponent systems.



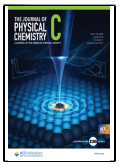
1. INTRODUCTION

With the popularization of healthy dietary concepts, food safety has become a core concern for consumers. Olive oil, renowned for its rich content of monounsaturated fatty acids and antioxidants as well as its potential to improve blood glucose control, has emerged as a high-end edible oil with continuously growing global consumption—this not only benefits from its nutritional advantages but also stems from consumers' strong recognition of its food safety performance and reliable quality.[1−7] Among various types of olive oil, extra virgin olive oil (EVOO) stands at the pinnacle of quality in the grading system established by the International Olive Council (IOC). This superior status is justified by its low acidity (≤0.8%), physical cold-pressing extraction process that avoids chemical treatments, and distinctive organoleptic properties.[8,9] In 2022 alone, the total value of globally imported olive oil and its fractions exceeded 9.74 billion US dollars, reflecting the rapid growth of the olive oil market. However, the expansion of market scale and the complexity of supply chains have led to an increasingly prominent adulteration issue in EVOO. Unscrupulous producers often adulterate EVOO with low-cost vegetable oils (e.g., peanut oil, soybean oil) and falsify content labels to seek exorbitant profits, which seriously undermines consumer rights and disrupts market order.[10−13] Therefore, the development of rapid and accurate technologies for authenticating olive oil and quantifying adulteration levels holds significant importance for safeguarding food safety and facilitating effective market supervision.

Traditional detection methods such as gas chromatography−mass spectrometry (GC),[14] high-performance liquid chromatography (HPLC),[15] and nuclear magnetic resonance (NMR) spectroscopy[16] can achieve oil component analysis; however, they suffer from limitations including complex sample pretreatment, long detection cycles, and strong dependence on equipment. Raman spectroscopy, by virtue of its advantages of noninvasiveness, high specificity, rapid response, and no requirement for complex sample preparation, has exhibited significant potential in the field of oil analysis.[17−21] This technique can distinguish different oil components based on molecular vibrational fingerprint information, yet it still faces several challenges in complex mixed systems. Specifically, the spectral features of various components in the mixture often overlap; matrix effects interfere with signals, and the weak signal characteristics of low-concentration mixtures are difficult to detect accurately. These factors significantly reduce the recognition accuracy of traditional chemometric methods for complex samples.[22−25]

In recent years, deep learning has achieved breakthrough progress in the field of spectral analysis. One-dimensional

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convolutional neural networks (1D-CNNs), by autonomously extracting spectral features, have demonstrated superior performance compared to traditional models in quantitative studies across multiple fields.[26−28] However, existing methods are mostly limited to single mixed types or low-dimensional mixed systems. When confronted with the complexity of multicomponent and multiproportion adulteration in real-world scenarios, a large number of independent models must be constructed for stepwise analysis, resulting in low detection efficiency and insufficient generalization ability. The multitask learning (MTL) framework enables the simultaneous realization of qualitative classification and quantitative regression through shared feature layers and parallel task training. This not only enhances model robustness but also reduces parameter redundancy.[29−33] While this multitask learning framework has been validated for its superiority in processing complex mixed spectra for the analysis of chemical warfare agent simulants, it remains underexplored in the field of edible oil adulteration detection. Specifically, reliable detection outcomes for this task demand not only accurate qualitative identification of adulterants and quantitative determination of adulteration levels but also explicit elucidation of the model's underlying decision-making mechanisms—such as which key spectral features govern the differentiation between authentic and adulterated oils. Such interpretability is pivotal for enhancing the credibility of detection results and facilitating their practical implementation in regulatory scenarios. However, existing models in this domain primarily prioritize the optimization of predictive performance, without in-depth investigation into model interpretability.[34]

In this study, a novel method for detecting EVOO adulteration based on Raman spectroscopy and a multiscale fusion convolutional residual attention network (MSF-CRAN) is proposed. To enhance the scientific credibility of the experiment, the SHapley Additive exPlanations (SHAP) algorithm is incorporated to conduct interpretability analysis of the model,[35] as it provides theoretically grounded feature-level attribution, enabling quantitative linkage between model predictions and specific Raman bands. To address the bottleneck of data scarcity in multicomponent adulteration systems in current research, an innovative strategy combining experimental construction and algorithmic simulation was adopted. First, binary and ternary adulterated olive oil samples—defined as mixtures of EVOO with one and two different adulterant oils, respectively, were prepared by blending pure olive oil with lower-cost edible oils (e.g., peanut oil, soybean oil, and corn oil). Higher-order adulteration systems involving three or more adulterant components were subsequently constructed to further increase compositional complexity. Quaternary and quinary blended data sets were generated via the spectral feature transfer algorithm to verify model performance. Simultaneously, a generative adversarial network (GAN) model was introduced as a data augmentation technique to optimize the model's generalization ability.[36,37] Experimental results demonstrate that by accurately capturing the source of discriminative spectral features between olive oil and the four types of vegetable oils, this model outperforms traditional deep learning algorithms (e.g., convolutional neural networks (CNNs) and U-net) under the tested conditions, achieving a classification accuracy of 100% for their differentiation and a ternary quantitative performance with a coefficient of determination (R^2) of 0.991 and a mean absolute error (MAE) of 0.0181. This study provides technical support for the future realization of on-site rapid multicomponent oil analysis.

2. MATERIALS AND METHODS

2.1. Materials

In this study, five types of edible vegetable oils were selected as experimental samples, specifically including extra virgin olive oil (brand: Oleo Bella, origin: Spain), soybean oil (brand: Jinfengsheng, origin: China), corn oil (brand: Arawana, origin: China), peanut oil (brand: Gujihua, origin: China), and rapeseed oil (brand: Tianfu Rapeseed Oil, origin: China). These edible vegetable oils originated from China and Spain and were purchased from Tmall Mall, ensuring the traceability of sample sources and their representativeness in the commercial market.

2.2. Sample Preparation

In this study, a library of binary and ternary adulterated olive oil samples was constructed by systematically blending olive oil with soybean oil, corn oil, peanut oil, and rapeseed oil, respectively. The specific preparation protocols are detailed as follows:

Preparation of binary adulterated samples: four binary adulteration systems were established using a pairwise blending mode, namely, olive oil−soybean oil, olive oil−corn oil, olive oil−peanut oil, and olive oil−rapeseed oil. For each system, 9 volume concentration gradients were set: the volume fraction of olive oil was 10, 20, 30, 40, 50, 60, 70, 80, and 90% in sequence, with the corresponding volume fraction of the adulterant vegetable oil decreasing from 90 to 10% at the same gradient interval.

Preparation of ternary adulterated samples: three ternary adulteration systems were constructed using a three-component blending mode, namely, olive oil−soybean oil−corn oil, olive oil−corn oil−peanut oil, and olive oil−peanut oil−rapeseed oil. Each system was designed with 9 volume concentration gradients: the volume fraction of olive oil in each gradient was 10, 20, 30, 40, 50, 60, 70, 80, and 90% sequentially, while the remaining volume was equally distributed between the other two oils.

All sample preparation processes were conducted at room temperature (25 ± 1 °C). Each component was accurately measured using a pipet and then transferred into a 5 mL centrifuge tube. Immediately after preparation, the samples were sealed and stored in the dark for subsequent testing.

2.3. Spectral Data Acquisition

Raman spectra of the samples were acquired using a confocal micro-Raman spectrometer (In Via Reflex, Renishaw plc, U.K.). A semiconductor laser with a wavelength of 532 nm was used as the excitation source. The 532 nm laser of the Renishaw in Via Reflex system has a maximum output power of 50 mW, and the laser power was set to 10% of the maximum output, corresponding to an absolute power of approximately 5 mW for the sample. The single-spectrum integration time was set to 10 s. This power level was selected to ensure an adequate signal-to-noise ratio (SNR) while minimizing potential local heating effects in the liquid-phase olive oil samples. For spectral acquisition, a grating with 1800 grooves per millimeter (1800 grooves/mm) and a 50× objective lens were employed. Prior to acquisition, the samples were observed in real time via a microscope, and the objective lens focal length was adjusted precisely to obtain optimal optical images, ensuring the highest quality of the spectral signals. Based on the typical research range of characteristic spectra for edible oils, the wavenumber range for spectral acquisition was set to 200−2000 cm^{-1}.

Data repeatability: to ensure the representativeness and statistical reliability of the data, ten sets of Raman spectra were collected for each sample.

Spectral preprocessing: all acquired raw spectral data were processed using LabSpec 6 spectral analysis software. The data processing workflow included normalization and smoothing.

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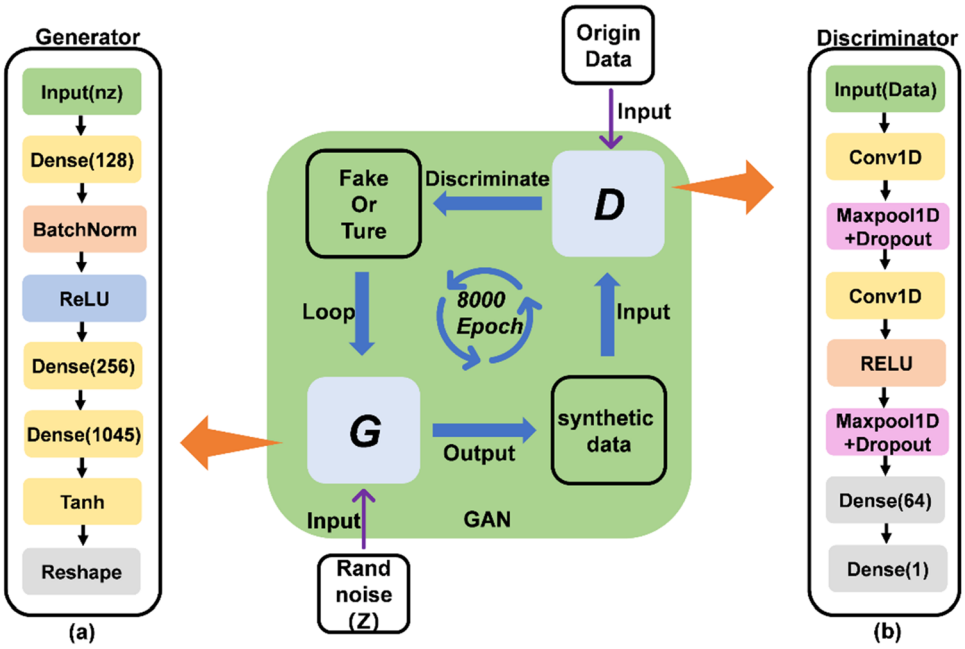


Figure 1. Schematic diagram of the GAN network structure. (a) Schematic diagram of the generator structure; (b) schematic diagram of the discriminator structure.

2.4. Data Augmentation

Deep learning models, particularly complex CNN, typically require large-scale, high-quality labeled data sets to achieve sufficient training, reach optimal performance, and avoid overfitting. However, in the field of Raman spectral analysis—especially for specific applications such as adulterated olive oil detection, acquiring a large number of labeled samples (especially those covering different concentration gradients) is often costly and time-consuming. This has become one of the primary bottlenecks limiting the improvement of model performance. To address the issue of insufficient training data, this study introduced generative adversarial network (GAN) technology to effectively augment the original Raman spectral data set.[38]

A GAN is a framework composed of two components: a generator and a discriminator. It achieves adversarial training by having the generator generate fake data, and the discriminator judges the authenticity of this data. The generator maps a random noise vector to fake spectral data with a distribution similar to that of real samples through a series of transposed convolution operations. In contrast, the discriminator performs feature extraction and binary classification (to distinguish between real and fake samples) on the input real or generated samples via convolutional layers. During the training process, the generator and discriminator continuously optimize their parameters through adversarial gameplay: the generator strives to generate samples realistic enough to deceive the discriminator, while the discriminator constantly improves its ability to identify real versus fake samples. Eventually, a Nash equilibrium is reached, at which point the generator can output synthetic samples that closely match the distribution of real spectral data.

In this study, for each adulterated olive oil sample with a specific concentration gradient, 10 raw spectral data points were input into the GAN model as the training set. After 8000 iterations of training, the generator stably outputs high-quality synthetic spectra. Eventually, the number of spectral data for each sample was expanded from 10 to 100, which effectively increased the data set size and provided sufficient sample support for the subsequent training of deep learning models. The specific network structure of the GAN model is illustrated in Figure 1, while the process of GAN network data training is presented in Figures S1 and S2.

2.4.1. Analysis of the GAN Network Structure. In this study, the generator architecture comprises three main modules: first is the fully connected layer module, which takes a noise vector as input, gradually expands its dimensionality, and generates a relatively rich feature representation. Second is the feature refinement module, which utilizes batch normalization layers and Rectified Linear Unit (ReLU) activation functions to enhance feature expression capability, ensure training stability, and further extract high-order features of the data. Third is the output layer, which generates the final feature map and constrains its output via a tanh activation function; this ensures the generated results fall within the range of $[-1, 1]$, thereby meeting the data normalization requirements. Through the layer-by-layer enhancement of feature dimensionality and nonlinear mapping, each module enables the generator to effectively capture the data distribution.

The discriminator adopts a convolutional neural network (CNN) architecture, extracting multilevel features from the input spectra through a progressive downsampling approach. First, two convolutional layers are used to extract local features, and a max-pooling layer is employed for dimensionality reduction; this enhances the discriminator's sensitivity to features of different scales. Next, dropout technology is applied to mitigate overfitting, and fully connected layers are introduced to classify the features. During the forward propagation process, the discriminator gradually reduces the size of the feature maps through two convolutional modules and max-pooling operations. Finally, the features are flattened and fed into the fully connected layers to perform the binary classification task. This design enables the discriminator to effectively distinguish between real and generated spectra based on multilayer features.

The loss function of a GAN typically consists of two components: the generator loss and the discriminator loss.

The goal of the discriminator is to maximize its output for real data (approximately 1) and minimize its output for generated data (approaching 0). Its loss function is expressed as follows

$$L_D = -E_{x\sim p_{\text{data}}}[\log D(x)] - E_{z\sim p_z}[\log(1 - D(G(z)))] \quad (1)$$

where p_{data} denotes the real data distribution, p_z represents the noise distribution, and $G(z)$ is the output of the generator.

The goal of the generator is to maximize the probability that the discriminator mistakenly classifies generated samples as real. Its loss function is expressed as follows

$$L_G = -E_{z\sim p_z}[\log(1 - D(G(z)))] \quad (2)$$

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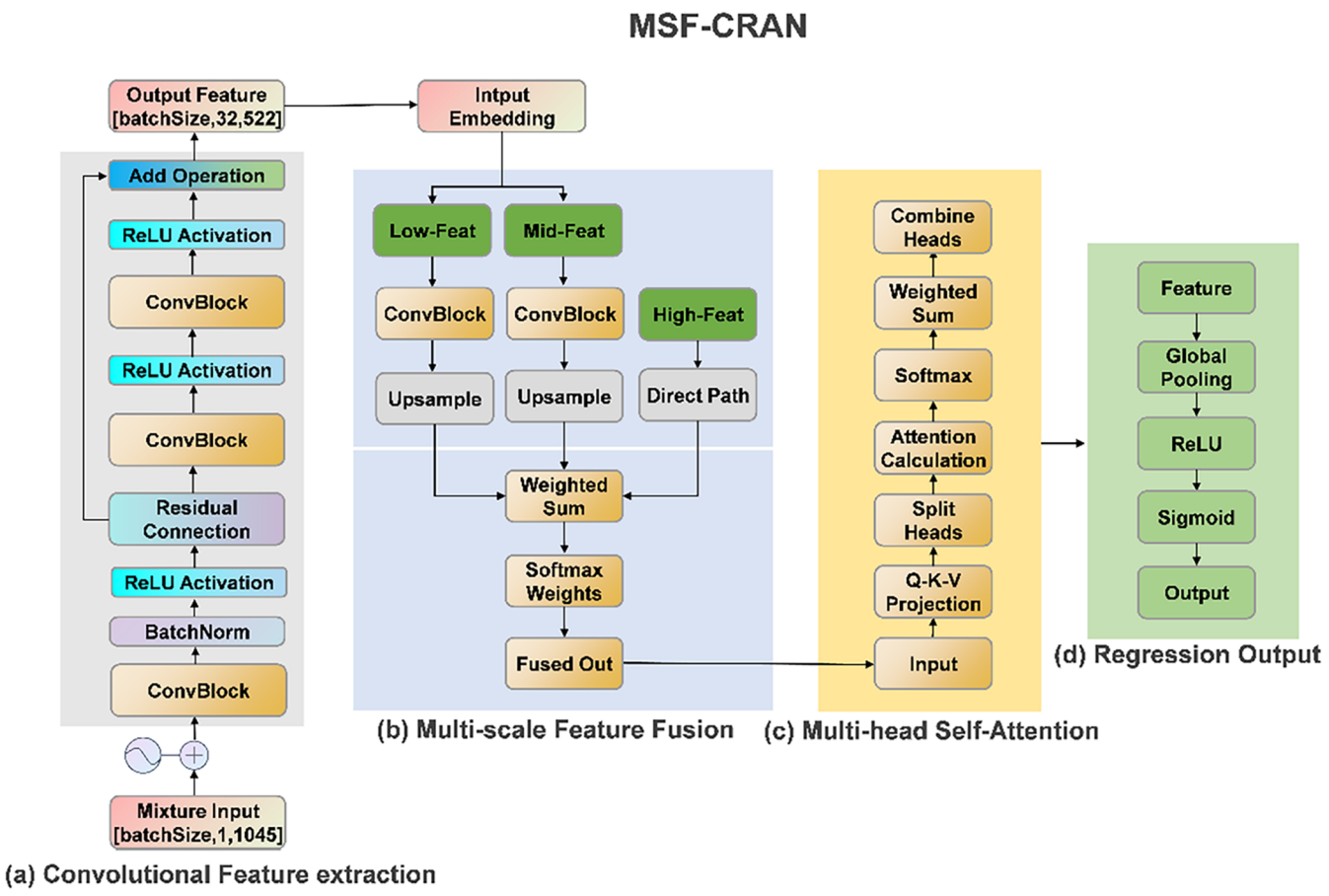


Figure 2. Schematic diagram of the qualitative and quantitative model. (a) Convolutional feature extraction module. (b) Multiscale feature fusion module. (c) Spectral self-attention module. (d) Regression output module.

By continuously optimizing this loss function, the generator drives $D(G(z))$ to approach 1, thereby successfully deceiving the discriminator.

The formulas for the generator and discriminator models facilitate a clear description of the operations at each network layer: the generator generates samples via fully connected layers and nonlinear activation functions, while the discriminator extracts features through convolutional layers and distinguishes the authenticity of samples. Ultimately, by optimizing the loss functions of both the generator and the discriminator, the generator can produce samples that closely resemble real data.

2.5. Network Framework for Qualitative and Quantitative Analysis

The network architecture used in this study is illustrated in Figure 2. The MSF-CRAN framework primarily consists of four components: a convolutional feature extraction module, a multiscale feature fusion module, a spectral self-attention module, and a regression output module.

2.5.1. Convolutional Feature Extraction Module. To effectively capture feature information at different scales in Raman spectra, this module is designed as a three-stage progressive convolutional neural network.

The primary convolutional layer employs a large kernel size (kernel size = 7) to capture the global contour features of the spectrum (e.g., baseline drift and broad spectral peaks). The intermediate convolutional layer uses a medium-sized kernel (kernel size of 5) to identify characteristic peak regions. In contrast, the deep convolutional layer adopts small kernels (kernel size = 3) to resolve fine peak structures. A residual connection block is incorporated after each stage of convolution; this block consists of layer normalization and two convolutional layers, which can effectively preserve the original spectral features and alleviate the gradient vanishing problem in deep networks. This hierarchical design enables adaptive processing of experimental spectra with varying signal-to-noise ratios (SNRs), realizing multiscale feature extraction from macro to micro levels.

2.5.2. Multiscale Feature Fusion Module. To address the significant difference in feature scales among different chemical components in mixed systems, this module proposes a cross-level feature fusion mechanism.[39] First, 1 × 1 convolutions are used to project the output feature maps of the shallow layer (capturing broad peak features) and intermediate layer (capturing medium-scale features) to the same channel dimension as the deep layer features (capturing fine structures). Second, bilinear interpolation is employed to unify the spatial dimensions of the feature maps from each layer to the scale of the deep layer feature maps. Finally, learnable weight parameters (normalized via softmax) are introduced to automatically adjust the contribution ratio of features from each layer, forming a fused multiscale feature representation.

The weighted fusion formula is expressed as follows

$$F_{\text{fused}} = \sum_{i=1}^{3} \alpha_i F_i \tag{3}$$

where F_{fused} denotes the fused feature, F_i represents the one-dimensional feature vector of the i-th scale (i = 1, 2, 3), and α_i is the learnable weight corresponding to the i-th scale feature vector, which is calculated using the softmax function.

Through the softmax function, the calculation formula for weight α_i remains unchanged

$$\alpha_i = \frac{e^{w_i}}{\sum_j^3 e^{w_i}} \tag{4}$$

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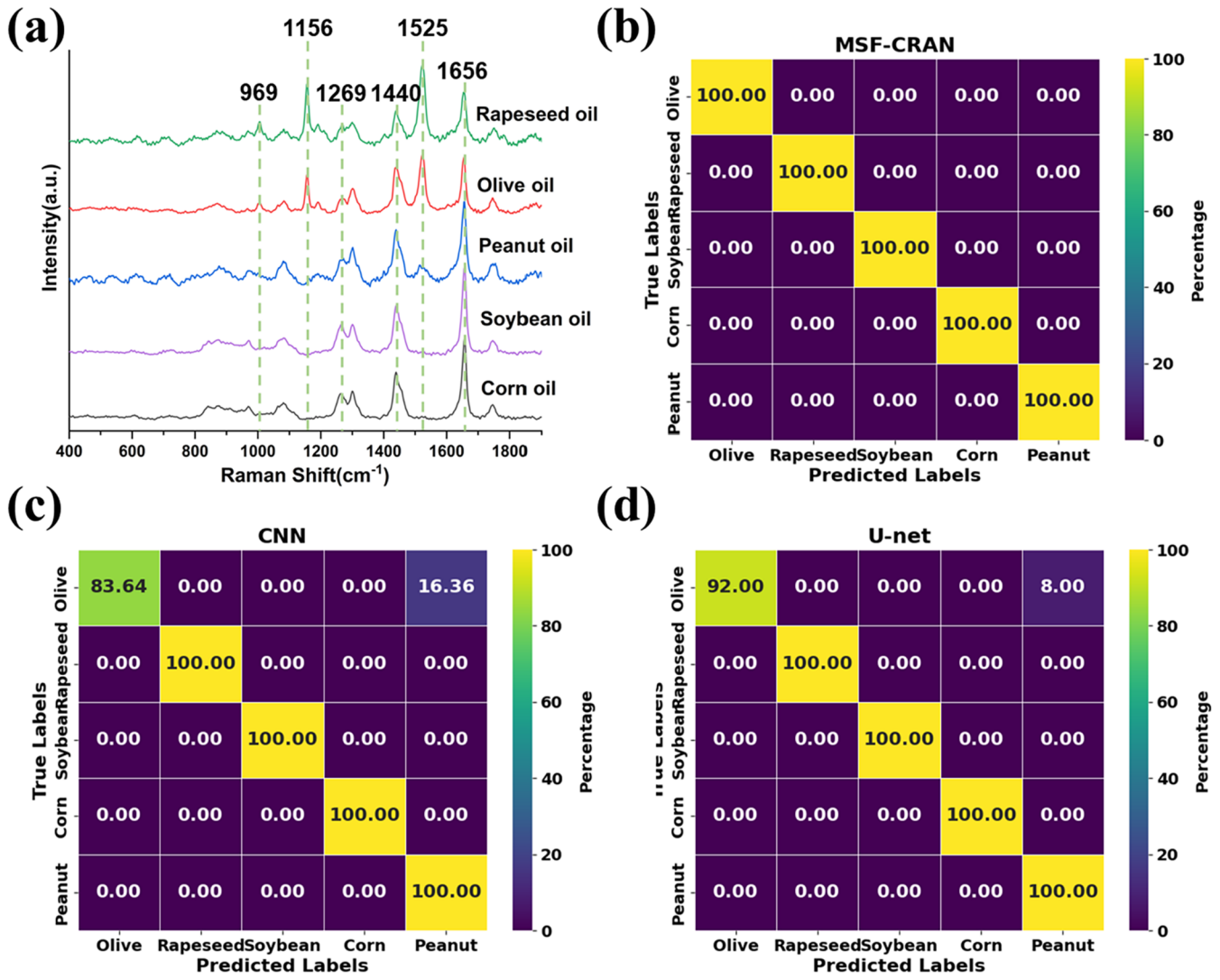


Figure 3. Classification of different vegetable oils using machine learning models. (a) Preprocessed Raman spectra of five types of vegetable oils. (b−d) Confusion matrices for the classification of vegetable oil spectral data using the MSF-CRAN, U-net, and CNN models, respectively.

where w_i is the learnable parameter of the i-th scale and $\sum_j^3 e^{w_i}$ is the normalization factor for the learnable parameters of all scales, ensuring the sum of all weights equals 1.

In this way, the model can adaptively adjust the weights of the feature vectors at different scales. The fused feature vector F_{fused} will contain important information from multiple scales, which is used for subsequent analysis or prediction tasks.

2.5.3. Spectral Self-Attention Module. To establish long-range dependencies between characteristic peaks in Raman spectra, the spectral self-attention module incorporates a multihead attention mechanism. This mechanism maps the spectral feature sequence to query vectors (Query), key vectors (Key), and value vectors (Value), respectively. The correlation weight matrix between spectral points (i.e., the attention matrix) is obtained by calculating the dot product of query vectors and key vectors, which is then used for the weighted aggregation of value vectors. This process can automatically focus on characteristic peak regions and suppress the noise interference. Notably, the multihead design (four heads adopted in this model) enables the model to simultaneously attend to the vibration modes of different chemical bonds, effectively capturing the characteristic peak-coupling effect of each component in the mixed system and significantly improving the detection sensitivity for low-concentration components.[40]

In the multihead attention mechanism, the calculation formula for each head is expressed as follows

$$\text{attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_{\mathrm{k}}}}\right)V \tag{5}$$

where Q denotes the query vector, K represents the key vector, V is the value vector, and d_{k} is the dimension of the key vector. By computing attention for different parts of the features, the multihead attention mechanism can capture global relationships in spectral data, thereby enhancing the understanding of details in oil samples. Particularly, in adulterated olive oil analysis, it helps the network identify subtle differences between different components.

2.5.4. Regression Output Module. First, this module compresses the spatial dimension of spectral data via global average pooling to extract feature vectors with a global statistical significance. Subsequently, a two-layer fully connected network is employed to realize the nonlinear mapping from feature vectors to concentration values: the first layer (64 neurons) performs high-dimensional feature transformation and applies the ReLU activation function, while the second layer (number of neurons equal to the number of components) outputs normalized concentration predictions. A dropout layer (dropout rate of 0.2) is incorporated into the network to prevent overfitting. Finally, the Sigmoid activation function is used to constrain the output within the range of [0, 1], which corresponds to the relative concentration percentage of each component. This design ensures the efficiency of feature utilization while enhancing the

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generalization ability of the model, making it particularly suitable for quantitative analysis tasks under the condition of limited samples.

2.6. Simulated Mixed Data

To comprehensively evaluate the generalization capability and quantitative accuracy of the proposed model under complex adulteration scenarios, this study employed a spectral mixing simulation algorithm for constructing high-dimensional adulteration system data sets (quaternary and quinary mixtures) based on experimentally prepared binary and ternary mixed samples. To complement the data set, a GAN-based data augmentation method was concurrently employed to generate realistic binary and ternary adulterated samples to augment the basic experimental data set—in contrast to the latter, which is dedicated to high-dimensional system construction. Given the challenges of high cost, cumbersome procedures, and incomplete coverage of all possible combinations in experimentally preparing real mixed samples with multiple components and high proportions, computational simulation has become a necessary and efficient supplementary strategy.

Principally, the GAN-based data augmentation method acquires the spectral distribution features of binary and ternary adulterated samples through the adversarial training of the generator and discriminator, focusing on simulating spectral details and distribution patterns of real samples to produce highly realistic new samples. In contrast, the adopted spectral mixing simulation algorithm, based on the linear/nonlinear superposition principle of spectral signals in multicomponent systems, directly constructs high-dimensional adulteration scenario data sets by quantitatively simulating the spectral responses of quaternary and quinary adulterated systems at different ratios. A brief description of the quaternary blended data simulation process is provided herein.

Assumed to be C_1, C_2, C_3, and C_4, which satisfy the following relationship

$$C_1 + C_2 + C_3 + C_4 = 1 \tag{6}$$

Subsequently, a spectral mixing simulation was performed on the spectral data. The spectral information on each component was mixed via a weighted synthesis method, which accounts for the contribution of each component's concentration to the final spectrum. A nonlinear response factor was also applied to simulate real-world conditions, as expressed in the following formula

$$S_{\text{mixed}} = \sum_{i=1}^{4} C_i(1 - 0.05C_i)S_i \tag{7}$$

Considering the interactions between different components, the interaction effects between all possible compound pairs were incorporated into the mixed spectrum, described by the formula

$$I_{ij} = \frac{0.02 \cdot C_i C_j (S_i S_j)}{\max(S_i) \cdot 4} \tag{8}$$

where I_{ij} represents the interaction effect between various components; 0.02 denotes the interaction effect coefficient, which controls the intensity of the interaction effect; $\max(S_i)$ is the maximum value of the spectrum of component i, used for normalization; and 4 is an additional normalization factor to further reduce the influence of interaction effects.

Combining the basic mixing process and interaction effects, the formula for the final mixed spectrum is

$$S_{\text{final}} = \sum_{i=1}^{4} c_i(1 - 0.05c_i)S_i + \sum_{i<j} I_{ij}M_{ij}0.5 \tag{9}$$

where S_{final} is the final mixed spectrum and 0.5 is a coefficient that halves the intensity of the interaction effect.

The simulation process for quinary blended data is identical to that for quaternary blended data. This process simulates the nonlinear characteristics of spectral mixing in real-world scenarios, accounts for the interactions between compounds, and employs smoothing to make the generated spectra more consistent with the actual measurement results.

3. RESULTS AND DISCUSSION

3.1. Raman Spectral Analysis of Edible Oils

Figure 3a presents the preprocessed Raman spectra of five types of edible vegetable oils. The Raman peaks of edible vegetable oils are mainly concentrated in the range of 700−1800 cm^{-1}.[18] In this study, qualitative and quantitative analyses of the chemical components of the five vegetable oils were conducted using Raman spectroscopy. Different Raman absorption peaks correspond to the characteristic vibrations of various chemical bonds in the vegetable oils. These vibration modes are closely related to the fatty acid composition, carotenoid content, and other chemical components of the vegetable oils.

Table 1 summarizes the assignment of major characteristic peaks in these oils and presents the chemical bonds and chemical sources corresponding to each Raman peak. Different absorption peaks in Raman spectra reflect the characteristic chemical bonds in vegetable oils, with each peak representing a specific chemical vibration mode. The Raman peak at 969 cm^{-1} corresponds to C═C bending vibration, which is mainly derived from unsaturated fatty acids.[41] This peak typically appears in unsaturated fatty acids containing double bonds, especially in fatty acid structures with conjugated double bonds, such as linoleic acid and linolenic acid. The peak at 1080 cm^{-1} is attributed to C−C stretching vibration, which is associated with fatty acids and reflects the degree of unsaturation of vegetable oils. The peak at 1156 cm^{-1} corresponds to C−C stretching vibration, related to carotenoids.[42] Carotenoid molecules contain multiple conjugated carbon−carbon single bonds, and this structural feature leads to the appearance of this peak. The peak at 1269 cm^{-1} is assigned to the C−H bending vibration, originating from olefinic compounds. This peak usually occurs in fatty acids with olefin structures or other unsaturated lipids. The Raman peak at 1300 cm^{-1} is the C−H bending twist vibration, with its chemical source being saturated hydrocarbons. This vibration mode is common in saturated fatty acids or hydrocarbon molecules and is particularly prominent in longer carbon chain structures. The peak at 1440 cm^{-1} corresponds to the C−H scissoring vibration, mainly derived from saturated hydrocarbons. The $-CH_2$ groups in saturated fatty acids typically exhibit significant vibration characteristics at this frequency, reflecting the saturated carbon chain structure of the fatty acids. The peak at 1525 cm^{-1} is attributed to C═C stretching vibration, primarily originating

Table 1. Assignment of Characteristic Peaks in Vegetable Oil Raman Spectra

Raman peak	attribution	chemical source
969 cm^{-1}	C═C bending	unsaturated fatty acids[41]
1080 cm^{-1}	C−C stretching	fatty acids[41]
1156 cm^{-1}	C−C stretching	carotenoids[42]
1269 cm^{-1}	C−H bending	alkene[18]
1300 cm^{-1}	C−H bending twist	saturated hydrocarbons[18]
1440 cm^{-1}	C−H scissoring	saturated hydrocarbons[41]
1525 cm^{-1}	C═C stretching	carotenoids[42]
1656 cm^{-1}	C═C stretching	oleic acid[41]
1742 cm^{-1}	C═O stretching	ester bond[43]

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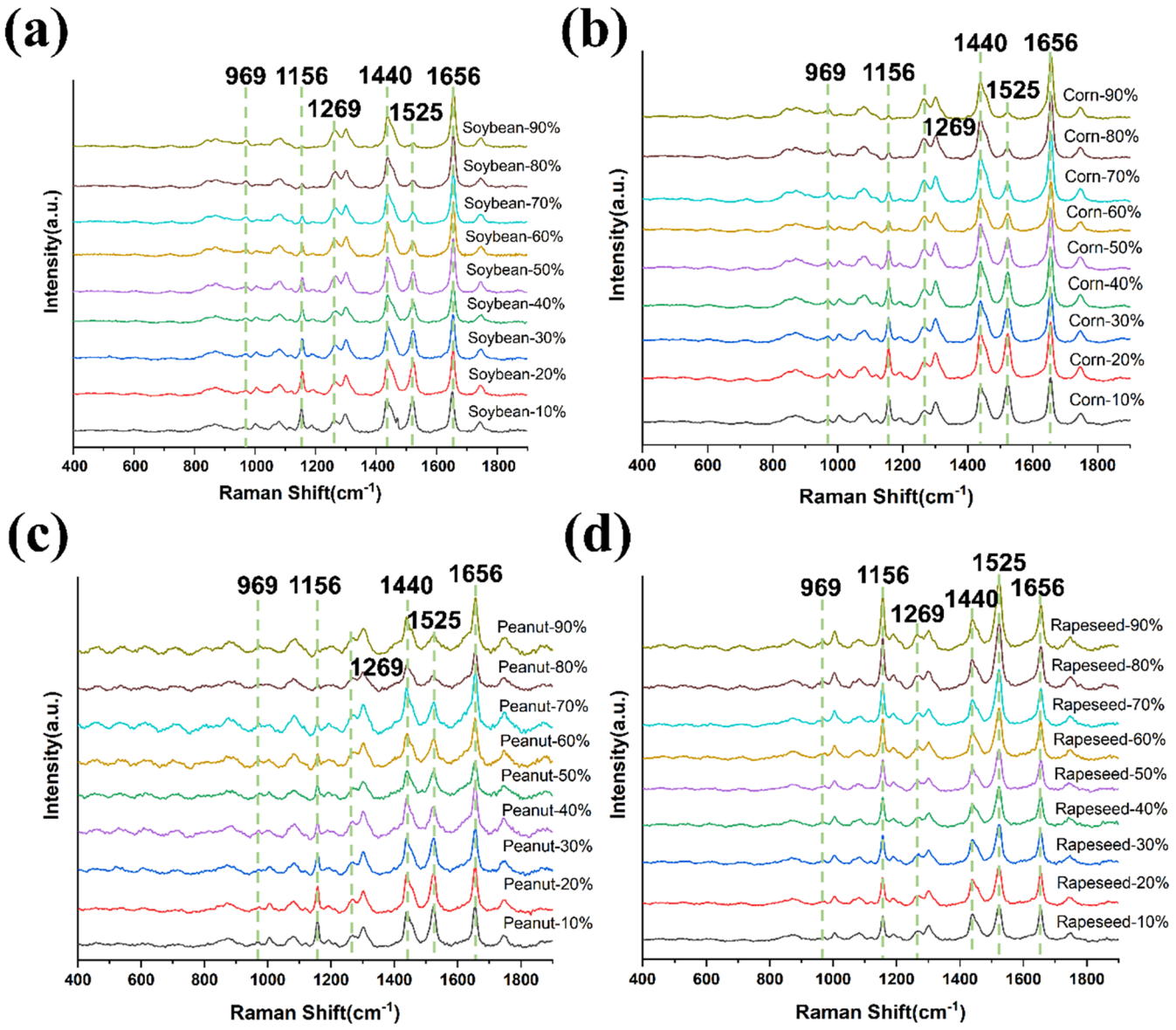


Figure 4. Presents gradient Raman spectral data of binary mixed systems formed by olive oil and different types of vegetable oils. (a–d) Correspond to the Raman spectra of binary mixed samples: soybean oil–olive oil, corn oil–olive oil, peanut oil–olive oil, and rapeseed oil–olive oil.

from the conjugated double bond structure of carotenoids, and is generally associated with the vibration of C=C bonds in the multiple conjugated structures of carotenoids.[42] The Raman peak at 1656 cm^{-1} corresponds to the C=C stretching vibration, with its chemical source being oleic acid. Oleic acid molecules contain a single C=C double bond; this feature makes the peak clearly visible in Raman spectra and is often used to identify vegetable oils containing unsaturated fatty acids. Finally, the peak at 1742 cm^{-1} is assigned to C=O stretching vibration, with its chemical source being ester bonds—especially the triglyceride structure in vegetable oils. This absorption peak indicates the vibration characteristics of ester bonds in vegetable oils.[43]

The analysis of these Raman peaks provides spectral characteristics of the main chemical components in vegetable oils, laying a theoretical foundation for the subsequent qualitative and quantitative analysis of vegetable oil Raman spectra using machine learning.

3.2. Classification of Different Vegetable Oils

To verify the analytical capability of the MSF-CRAN network and evaluate its generalization performance, this study employed a model for qualitative analysis of olive oil and the other four types of vegetable oils.

3.2.1. Data Analysis. To achieve an accurate analysis of vegetable oil spectral data, this study employed the MSF-CRAN network for classification. This network structure exhibits superior feature extraction capability, enabling more accurate identification and differentiation of various vegetable oil categories, and demonstrates excellent performance in qualitative analysis.

Therefore, this study further utilized the MSF-CRAN network to classify vegetable oil spectral data and verified its performance in qualitative analysis using another two traditional deep learning models.

In practical measurements, slight variations exist in the oil spectral data of each sample, manifested as fluctuations in peak

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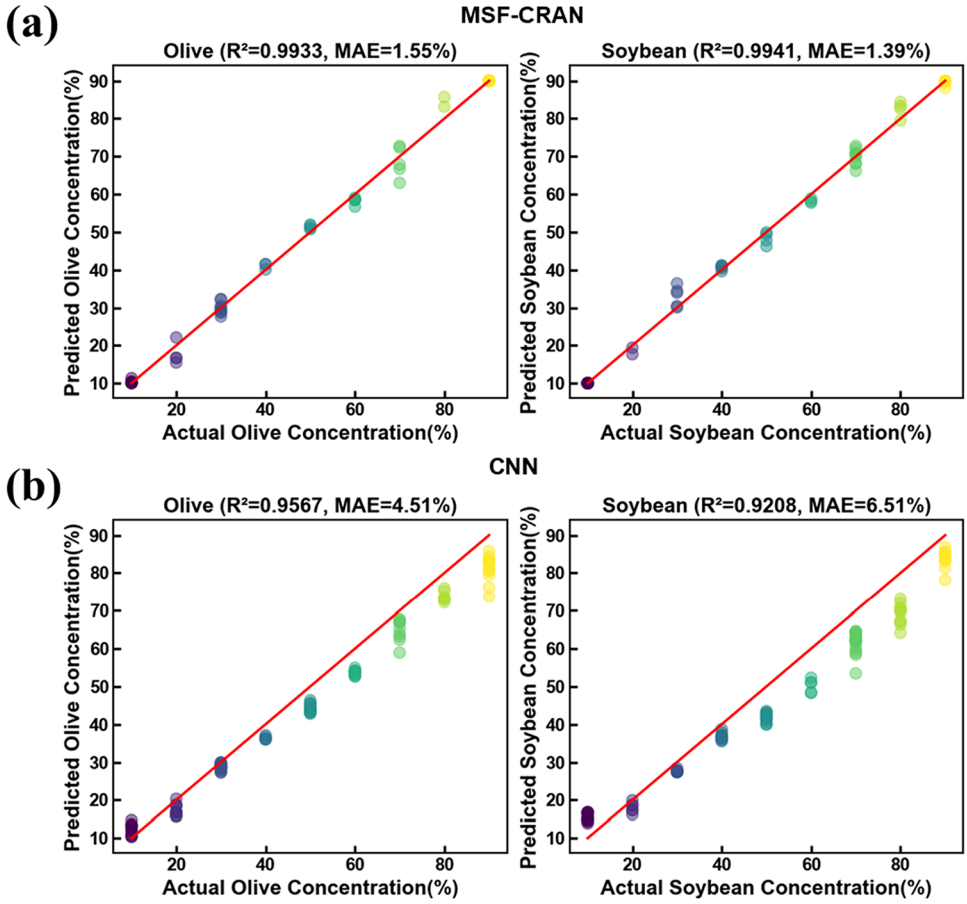


Figure 5. Quantitative results of two models for binary blended olive oil. (a) Scatter plot of predicted vs actual concentrations of olive oil and soybean oil by the MSF-CRAN model, showing excellent linear correlation (R^2 = 0.994). (b) Scatter plot of predicted vs actual concentrations of olive oil and soybean oil by the CNN model, demonstrating good prediction performance (R^2 = 0.957).

positions and intensities. To determine the minimum number of spectra required for classification while ensuring sufficient robustness, this study calculated the Pearson correlation coefficient (PCC) between the average spectra derived from two random subsets of the olive oil spectrum data set. The relationship plots between the number of spectra of other vegetable oils and PCC are presented in Figure S3. Analysis results show that when approximately 60 spectra are used for average calculations, the PCC curve converges rapidly to a value close to 1. In summary, to ensure that the final average spectra exhibit maximum robustness, this study employed a GAN to augment the spectral data of each sample category, ultimately increasing the number of spectra per sample category to 100.

The data set partitioning strategy was designed to ensure a clear separation between model training and performance evaluation. The simulated spectra generated by the GAN were primarily used to construct the training data set for the deep learning model. These GAN-generated data were derived from a limited number of experimentally measured Raman spectra and were intended to enrich the diversity of the training samples in multicomponent adulteration scenarios. The independently measured real vegetable oil spectra were reserved as the test set and were not involved in either GAN training or model parameter optimization. This design ensures that the model performance was evaluated exclusively on experimentally acquired data, thereby providing a realistic assessment of generalization capability and avoiding information leakage. During the training process, the training set was used for feature learning and weight adjustment. A validation subset was defined within the training data to monitor model performance during training and to support hyperparameter optimization. The test set was used only for the final performance evaluation after model training was completed.

This section presents the performance of three machine learning models in classifying the spectral data of five vegetable oils, specifically including the classification results of the MSF-CRAN, U-net,[44] and CNN[45] models. Figure 3b–d shows the confusion matrices for the classification of these three models, illustrating the classification performance and misclassification cases of each model for different vegetable oil categories.

Visual inspection of the figures reveals that the MSF-CRAN model exhibits excellent classification performance with an accuracy of 100%. This result indicates that the MSF-CRAN model can accurately distinguish the spectral data of the five vegetable oils without any misclassification. Its outstanding performance may be attributed to more refined adjustments during feature extraction and model optimization, enabling it to better capture subtle differences in oil spectra and, thus, achieve perfect classification. In contrast, the U-net and CNN models show slightly inferior performance with accuracies of approximately 97%. Although these two traditional machine learning algorithms can still classify vegetable oils effectively, they exhibit misclassification cases compared with the MSF-CRAN model. The U-net model may have certain limitations in the configuration of convolutional layers and network

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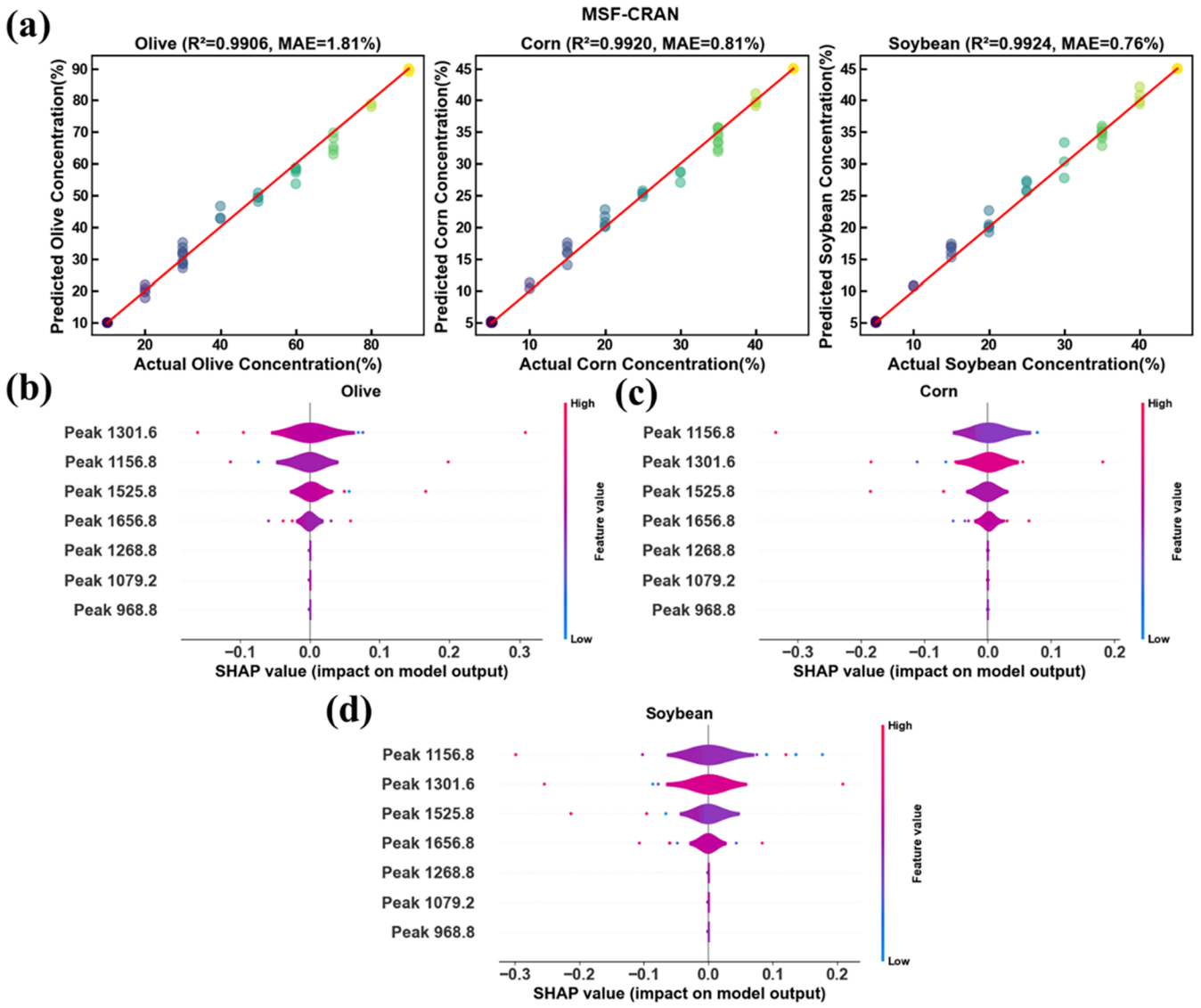


Figure 6. Quantitative results of ternary blended olive oil using the MSF-CRAN model. (a) Scatter plot of predicted vs actual concentrations of olive oil, corn oil, and soybean oil by the MSF-CRAN model, showing excellent linear correlation ($R^2 > 0.99$). (b–d) SHAP violin plots illustrating the contributions of several major characteristic peaks to the quantitative prediction of olive oil, soybean oil, and corn oil, respectively.

structure when processing spectral data, resulting in lower discriminability between some categories than for the MSF-CRAN model. While the CNN model is widely used in visual data processing, its feature extraction and classification for Raman spectral data are relatively simplistic and may fail to fully utilize the potential information in spectral data. In summary, although all three models can effectively classify vegetable oils, the MSF-CRAN model undoubtedly demonstrates the most superior performance in this study, highlighting its efficiency and robustness in Raman spectral data classification tasks.

3.3. Quantitative Analysis of Adulterated Olive Oil

In this section, soybean oil, peanut oil, rapeseed oil, and corn oil at different ratios were individually blended into olive oil to construct a series of adulterated samples.

3.3.1. Analysis of Binary Blended Olive Oil Spectral Data Set. As illustrated in Figure 4a, binary mixed samples were prepared by blending soybean oil into olive oil at ratios of 10, 20, 30, 40, 50, 60, 70, 80, and 90%, respectively. As shown in the figure, the Raman spectra of the binary mixed samples with different adulteration concentrations exhibit high overall similarity with only minor differences at the macro level. For instance, significant changes in spectral intensity can be observed in the local regions of the key characteristic peaks at 1156, 1269, and 1656 cm^{-1}. Figure 4b–d depicts the Raman spectra of three other binary adulterated mixtures (corn oil–olive oil, peanut oil–olive oil, and rapeseed oil–olive oil), respectively, all of which exhibit high similarity and are not readily distinguishable by the naked eye.

3.3.2. Quantitative Analysis of Binary Blended Olive Oil. Based on the experimental results, the performance of three models (MSF-CRAN, U-net, and CNN) in the quantitative regression of binary blended olive oil was analyzed. For the quantification of binary blended olive oil, 900 augmented data points generated by the GAN network served as the training set, while 90 measured real blended data points were used as the test set.

As shown in Figure 5a, the MSF-CRAN model exhibits excellent linear correlation in predicting the concentrations of

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Table 2. Comparison of Validation Results for Blended Oil Datasets

combination dimensions		accuracy/R^2					
		GAN and random baseline noise			no GAN and random baseline noise		
		CNN	U-net	MSF-CRAN	CNN	U-net	MSF-CRAN
qualitative	pure	0.975	0.979	1.000	0.833	0.800	0.900
quantitative	binary	0.924	0.887	0.993	0.814	0.739	0.879
	ternary	0.887	0.891	0.991	0.761	0.439	0.843
	quaternary	0.827	0.907	0.973	0.603	F	0.834
	quinary	0.816	0.855	0.966	F	F	0.822

olive oil and soybean oil. For olive oil, the predicted coefficient of determination (R^2) reaches as high as 0.9933 with an MAE of 0.0155, indicating that the model achieves extremely high accuracy in processing spectral data of binary blended oils. The CNN model shows relatively good performance, with an R^2 of 0.957 and an MAE of 0.0451 for olive oil (Figure 5b); although inferior to MSF-CRAN, it still enables effective prediction of oil concentrations. In contrast, the U-net model demonstrates moderate predictive performance, with an R^2 of 0.904 and an MAE of 0.0651 for olive oil (Figure S4). While it can perform valid predictions, its predictive accuracy is far inferior to that of MSF-CRAN. Figure S5 illustrates the training processes of the MSF-CRAN, U-net, and CNN models, respectively. The variation trends of their loss values are similar to those of MSF-CRAN, indicating that all three models exhibit a certain degree of convergence during training.

Overall, although all three models can effectively accomplish the task, the MSF-CRAN model demonstrates the most superior performance in terms of accuracy and stability, particularly showing higher performance in spectral data processing and oil concentration prediction.

3.3.3. Quantitative Analysis of Ternary Blended Olive Oil. This section presents the quantitative analysis of ternary blended olive oil by using the MSF-CRAN model. Nine concentration gradients are included with 100 synthetic spectral data points generated for each concentration. The data set partitioning follows the same method as that used for the quantitative analysis of binary blended olive oil: augmented data generated by the GAN serve as the training set, while the measured real data is used as the test set.

Figure 6a presents the scatter plot of predicted vs actual concentrations generated by the MSF-CRAN model. As clearly observed from this figure, an extremely strong linear correlation exists between the model's predicted results and the actual concentration values, with all R^2 values exceeding 0.99. This performance is consistent with the model's performance in the binary blending quantification task, further verifying the model's consistency and excellent robustness across different tasks. The results indicate that the MSF-CRAN model can effectively and accurately predict the concentrations of the three oils. Figure S6 illustrates the variation trend of loss values for the training set and validation set with the number of training epochs during the model training process. This figure reveals that the loss values gradually decrease as the model is optimized, demonstrating that the model gradually converges during training and can effectively avoid overfitting. Particularly, for the validation set loss, the model maintains an excellent generalization ability, which further confirms its reliability in practical applications.

Regarding model interpretability, Figure 6b–d displays SHAP violin plots for the three oil components, aiming to reveal the contributions of different features to the quantitative prediction of olive oil, soybean oil, and corn oil in the MSF-CRAN model. These plots illustrate the importance and influence of major characteristic peaks in the model's predictions. Specifically, the SHAP violin plot was constructed based on 100 randomly selected spectral data. In this plot, each row corresponds to one feature peak with the horizontal axis denoting SHAP values. Features are sorted by the mean absolute value of their SHAP values: the top-ranked features exert the most significant impact on the model output, while the bottom-ranked ones have a relatively weaker influence. Each data point in the SHAP violin plot corresponds to a single sample; a redder color indicates a higher Raman intensity of the feature peak, while a bluer color denotes a lower Raman intensity.

SHAP violin plots not only show the contribution distribution of each feature but also reveal the relationship between features and prediction results. From these plots, the features that play a more critical role in predicting the concentrations of the three oils can be clearly identified. For instance, the characteristic peak at 1269 cm^{-1} shows a greater contribution to quantitative prediction, while the peak at 969 cm^{-1} is relatively less important. These interpretable visualizations provide a clear perspective for understanding the model's decision-making process and also facilitate model optimization and improvement of its applicability across different scenarios. Figures S7 and S8 show the quantitative results of the MSF-CRAN model applied to another two ternary mixed olive oil systems.

3.4. Quantitative Detection of Olive Oil in Complex Systems

To further evaluate the model's performance in the quantitative analysis of quaternary and quinary blended olive oil, 15 quaternary blended olive oil samples and 9 quinary blended olive oil samples were constructed in this study. The spectra of the quaternary and quinary simulated blended samples are shown in Figures S9 and S10, respectively. The construction was based on the average spectra of pure vegetable oils, using a spectral mixing simulation algorithm with the introduction of minimal Gaussian noise interference. Each blended sample contained 50 spectral data points.

The quantitative results of the MSF-CRAN model were compared with those of the CNN and U-net models. The qualitative and quantitative analysis results of the three models for blended olive oil are presented in Table 2. Herein, "F" denotes values that are below the threshold or correspond to recognition errors.

Through comparative analysis of the data presented in Table 2, it is demonstrated that GAN-based data augmentation exerts a significant enhancing effect on model performance. In qualitative analysis (pure samples), after introducing GAN data augmentation, the classification accuracy of the CNN model

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increased from 0.833 to 0.975, that of U-net improved from 0.800 to 0.979, and that of MSF-CRAN rose from 0.900 to 1.000. All three models achieved remarkable performance improvements, among which MSF-CRAN attained a perfect classification accuracy after GAN-based augmentation. This indicates that data augmentation can effectively enhance the identifiability of features in pure samples.

In quantitative analysis, the gain effect of GAN augmentation becomes increasingly pronounced with the increase in the dimensionality of mixtures. For instance, in the quantitative analysis of ternary mixtures, the performance improvement of U-net is the most prominent, with its R^2 value soaring from 0.439 to 0.891. This indicates that GAN augmentation exhibits particularly remarkable efficacy in alleviating the issue of the insufficient feature extraction capability of this model in systems with moderate complexity. In the quantitative analysis of quinary mixtures, without GAN-based data augmentation, both CNN and U-net fail to yield valid results, while MSF-CRAN still maintains an R^2 value of 0.822. After introducing GAN augmentation, the R^2 values of CNN and U-net recover to 0.816 and 0.855, respectively, and that of MSF-CRAN is further enhanced to 0.966. This fully confirms that GAN augmentation can effectively expand the adaptability of models to complex systems.

Under identical experimental conditions, the MSF-CRAN model outperforms both CNN and U-net consistently across all mixture dimensionalities, particularly highlighting its superior feature extraction and matching capabilities in qualitative analyses (pure samples) and various quantitative analyses (i.e., binary, ternary, quaternary, and quinary mixtures). Specifically, MSF-CRAN achieves a classification accuracy of 1.000 for pure samples, which is significantly higher than that of CNN (0.975) and U-net (0.979). Furthermore, MSF-CRAN maintains high precision even with increasing dimensionality of mixtures; even in quinary mixtures, despite a slight decrease in precision, its accuracy of 0.966 still outperforms those of the other models.

Overall, by expanding the effective sample space, GAN significantly enhances the stability of all models in feature extraction and pattern matching. Particularly in high-dimensional mixed systems, it effectively mitigates performance degradation caused by insufficient information in the original data. Among all models, MSF-CRAN, when combined with GAN augmentation, exhibits the most outstanding performance. It not only achieves perfect classification in simple systems but also maintains a high goodness-of-fit of 0.966 in complex quinary mixtures, fully demonstrating the synergistic advantages of its network structure design and data augmentation strategy.

4. CONCLUSIONS

This study proposes an integrated framework for detecting adulteration in edible vegetable oils based on Raman spectroscopy and an MSF-CRAN. The methodological innovation of this work lies in the systematic integration of multiple strategies tailored to complex Raman spectral analysis. Specifically, (i) multiscale feature fusion is employed to capture discriminative information across Raman bands with varying widths and intensities; (ii) a unified multitask learning framework enables simultaneous qualitative classification and quantitative prediction of adulteration levels; (iii) SHAP-based interpretability analysis is incorporated to establish a direct linkage between model decisions and chemically meaningful spectral features, enhancing the transparency and chemical relevance of the model.

To address the issue of limited real samples in multicomponent adulteration systems, a hybrid data construction strategy combining experimental preparation and algorithmic simulation is developed. Binary and ternary mixed samples covering various adulteration concentrations are experimentally prepared, while spectral feature transfer technology is used to generate quaternary and quinary mixed spectral data sets, effectively expanding the training and validation space. In addition, GAN-based data augmentation further improves the generalization capability of the model.

Under the tested conditions, the constructed MSF-CRAN model achieves accurate quantitative prediction of adulteration content and demonstrates superior classification and regression performance in complex multicomponent scenarios compared with conventional machine learning approaches. Overall, this work provides not only a high-performance predictive tool but also a chemically interpretable analytical framework, offering a reliable technical solution for on-site rapid oil quality assessment and complex mixture analysis.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpcc.6c00601.

Additional figures including GAN training process, model loss curves, Pearson correlation analysis, quantitative prediction results, SHAP interpretation plots, and Raman spectra of multicomponent mixtures (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Jia-Sheng Lin − *College of Chemical Engineering and Materials Science, Quanzhou Normal University, Quanzhou 362000, China*; Email: jacksonlin.chem@gmail.com

Fan-Li Zhang − *College of Optical and Electronic Technology, China Jiliang University, Hangzhou 310018, China*; orcid.org/0000-0001-9396-7867; Email: Zhangfl@cjlu.edu.cn

Authors

Xue-Yang Xiong − *College of Optical and Electronic Technology, China Jiliang University, Hangzhou 310018, China*

Huan Chen − *College of Optical and Electronic Technology, China Jiliang University, Hangzhou 310018, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jpcc.6c00601

Notes

The authors declare no competing financial interest.

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